Plate Planner: An AI-Powered Graph-Enhanced Meal Planning and Recipe Recommendation System

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Abstract

Real-world meal planning requires balancing competing objectives that extend far beyond returning relevant recipes for a single query. Users must account for what ingredients are already available in their pantry, minimize the cost of acquiring new items, reduce food waste from unused perishables, and ensure that ingredient substitutions are feasible when exact matches are unavailable. Existing recipe retrieval systems predominantly optimize for single-recipe relevance, neglecting these week-level planning constraints. In this paper, we present PlatePlanner, a graph-augmented, pantry-aware semantic retrieval framework that selects multi-day meal plans while maximizing semantic relevance and minimizing incremental shopping burden and waste. Our architecture integrates a four-stage hybrid retrieval pipeline: (1) FAISS-based semantic search using SentenceTransformer embeddings over a corpus of 100K+ recipes, (2) Neo4j graph-based ontology filtering with context-aware ingredient substitution, (3) nutrient-aware mathematical ranking, and (4) retrieval-augmented generation (RAG) for explainable recommendations. Experimental evaluation demonstrates sub-100ms p95 latency, 92% user satisfaction for substitutions, 23% recall improvement over direct-only substitution, and 85% user preference for hybrid scoring.